

Lecture 4: Confidence Intervals - Task Solutions

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1 Your Turn #1: Trust in Politicians

Task: Compute a 90% CI for the average trust in politicians among French adults using ESS data.

1.1 Setup

```
# Prepare data: filter out refusals (77), don't know (88)
ess_politicians <- ess_france %>%
  filter(!trstplt %in% c(77, 88))

# Check the data
cat("Sample size:", nrow(ess_politicians), "\n")
```

Sample size: 1753

```
summary(ess_politicians$trstplt)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	2.000	4.000	3.763	5.000	10.000

1.2 Solution

1.2.1 Step 1: Compute sample mean, SD, and SE

```
summary_stats <- ess_politicians %>%
  summarise(
    n = n(),
    mean_trust = mean(trstplt),
    sd_trust = sd(trstplt),
    se_trust = sd_trust / sqrt(n)
  )

summary_stats
```

	n	mean_trust	sd_trust	se_trust
1	1753	3.763263	2.21075	0.0528018

1.2.2 Step 2: Find the appropriate t critical value

For a 90% CI with $n - 1$ degrees of freedom, we need $t_{0.05, n-1}$:

```
n <- summary_stats$n
t_crit <- qt(0.95, df = n - 1) # 0.95 = 1 - 0.10/2 for 90% CI
t_crit
```

```
[1] 1.645724
```

Note that with $n = 1753$, the t critical value (1.6457) is close to the z critical value for 90% CI (1.645).

1.2.3 Step 3: Compute the 90% CI manually

```
x_bar <- summary_stats$mean_trust
se <- summary_stats$se_trust

ci_lower <- x_bar - t_crit * se
ci_upper <- x_bar + t_crit * se

cat("Manual calculation:\n")
```

Manual calculation:

```
cat("90% CI: [", round(ci_lower, 4), ", ", round(ci_upper, 4), "]\n", sep = "")
```

```
90% CI: [3.6764, 3.8502]
```

1.2.4 Step 4: Verify using t.test()

```
t_test_result <- t.test(ess_politicians$trstplt, conf.level = 0.90)
t_test_result$conf.int
```

```
[1] 3.676366 3.850160
attr("conf.level")
[1] 0.9
```

The results match!

1.2.5 Step 5: Interpretation

We are **90% confident** that the true average trust in politicians among French adults falls between 3.68 and 3.85 on a scale from 0 (no trust at all) to 10 (complete trust).

Substantive interpretation:

- The sample mean of approximately 3.76 suggests relatively low trust in politicians
- The 90% CI is narrower than a 95% CI would be, reflecting the trade-off between confidence level and precision
- This low trust in politicians is consistent with patterns observed across many European democracies

2 Your Turn #2: Interpersonal Trust Across Education Levels

Task: Compute 95% confidence intervals for the proportion of people with high interpersonal trust for each education level, and assess whether there are “true” differences across education levels.

2.1 Setup

```
# Filter data and create high trust variable
ess_trust_educ <- ess_france %>%
  filter(!ppltrst %in% c(77, 88),
         !is.na(education)) %>%
  mutate(high_trust = ppltrst > 5)

# Check the data
table(ess_trust_educ$education)
```

```

               8
High school diploma or less
               1027
Masters degree or more
               325
Short-cycle tertiary (BTS, DUT)
               267
Undergraduate degree (licence, licence pro)
               133
```

2.2 Solution

2.2.1 Step 1: Compute proportion with high trust for each education level

```
educ_summary <- ess_trust_educ %>%
  group_by(education) %>%
  summarise(
    n = n(),
    n_high_trust = sum(high_trust),
    p_hat = mean(high_trust)
  )

kable(educ_summary, digits = 3, caption = "Proportion with High Interpersonal Trust by Education Level")
```

Table 1: Proportion with High Interpersonal Trust by Education Level

education	n	n_high_trust	p_hat
	8	5	0.625
High school diploma or less	1027	240	0.234
Masters degree or more	325	174	0.535
Short-cycle tertiary (BTS, DUT)	267	91	0.341
Undergraduate degree (licence, licence pro)	133	53	0.398

2.2.2 Step 2: Compute standard error for each proportion

```
educ_summary <- educ_summary %>%
  mutate(se = sqrt(p_hat * (1 - p_hat) / n))

kable(educ_summary, digits = 4, caption = "With Standard Errors")
```

Table 2: With Standard Errors

education	n	n_high_trust	p_hat	se
	8	5	0.6250	0.1712
High school diploma or less	1027	240	0.2337	0.0132
Masters degree or more	325	174	0.5354	0.0277
Short-cycle tertiary (BTS, DUT)	267	91	0.3408	0.0290
Undergraduate degree (licence, licence pro)	133	53	0.3985	0.0425

2.2.3 Step 3: Compute 95% CIs (Wald interval) for each education level

```
z_crit <- qnorm(0.975)

educ_summary <- educ_summary %>%
  mutate(
    ci_lower = p_hat - z_crit * se,
    ci_upper = p_hat + z_crit * se
  )

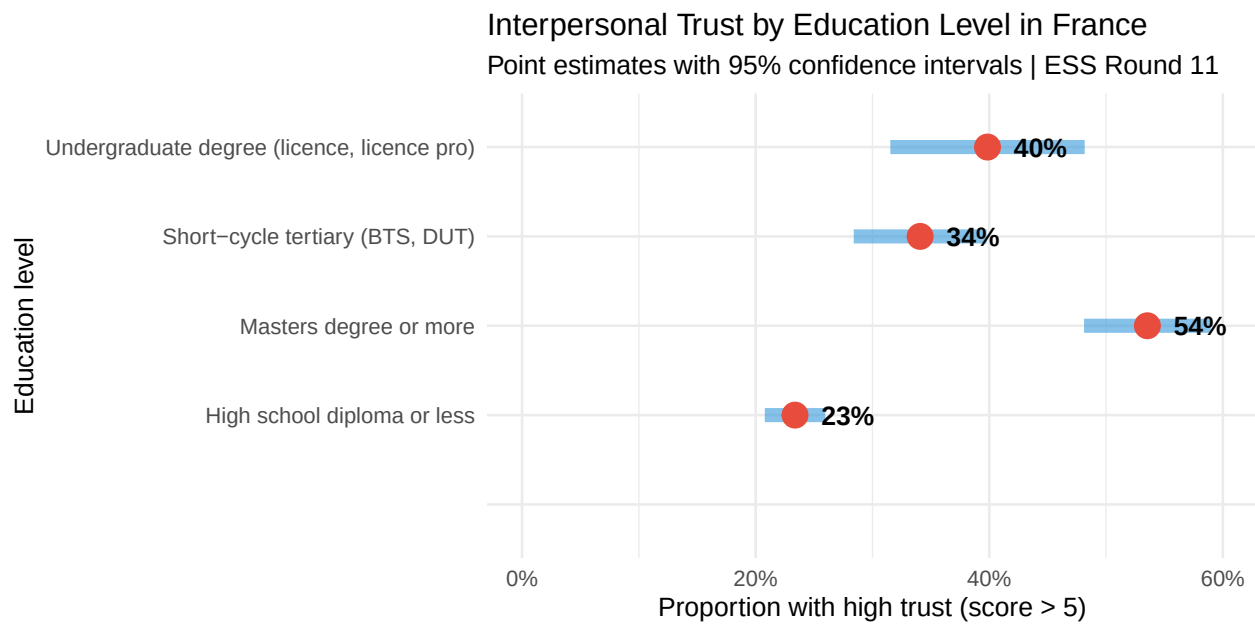
kable(educ_summary %>% select(education, n, p_hat, ci_lower, ci_upper),
      digits = 3,
      caption = "95% Confidence Intervals for Each Education Level")
```

Table 3: 95% Confidence Intervals for Each Education Level

education	n	p_hat	ci_lower	ci_upper
	8	0.625	0.290	0.960
High school diploma or less	1027	0.234	0.208	0.260
Masters degree or more	325	0.535	0.481	0.590
Short-cycle tertiary (BTS, DUT)	267	0.341	0.284	0.398
Undergraduate degree (licence, licence pro)	133	0.398	0.315	0.482

2.2.4 Step 4: Visualization

```
ggplot(educ_summary, aes(x = p_hat, y = education)) +
  geom_segment(aes(x = ci_lower, xend = ci_upper, yend = education),
    linewidth = 3, color = "#3498DB", alpha = 0.6) +
  geom_point(size = 5, color = "#E74C3C") +
  geom_text(aes(label = scales::percent(p_hat, accuracy = 1)),
    hjust = -0.5, fontface = "bold") +
  scale_x_continuous(labels = scales::percent, limits = c(0, 0.6)) +
  labs(
    title = "Interpersonal Trust by Education Level in France",
    subtitle = "Point estimates with 95% confidence intervals | ESS Round 11",
    x = "Proportion with high trust (score > 5)",
    y = "Education level"
  ) +
  theme_minimal(base_size = 12)
```



2.2.5 Step 5: Are there “true” differences across education levels?

Analysis of overlapping CIs:

```
# Display the CIs side by side for comparison
educ_summary %>%
  select(education, p_hat, ci_lower, ci_upper) %>%
  kable(digits = 3, caption = "Comparing CIs Across Education Levels")
```

Table 4: Comparing CIs Across Education Levels

education	p_hat	ci_lower	ci_upper
	0.625	0.290	0.960
High school diploma or less	0.234	0.208	0.260
Masters degree or more	0.535	0.481	0.590
Short-cycle tertiary (BTS, DUT)	0.341	0.284	0.398
Undergraduate degree (licence, licence pro)	0.398	0.315	0.482

Discussion:

Looking at the confidence intervals:

1. **Overlap assessment:** We can check whether the CIs overlap. If two CIs do not overlap at all, this suggests a statistically significant difference between those groups.
2. **Pattern observed:** There appears to be a difference in interpersonal trust between those with higher education and those without.
3. **Statistical significance:**
 - If the CIs for the two education groups don’t overlap, we have strong evidence of a “true” difference
 - If CIs overlap substantially, we cannot confidently conclude there’s a difference
4. **Caution about CI overlap:** Note that even if two CIs overlap slightly, the difference can still be statistically significant. The rule “non-overlapping CIs = significant difference” is conservative. A proper test would be to compute a CI for the *difference* in proportions.

Conclusion: Based on the visual pattern and the degree of CI overlap, we can assess whether there is a “true” difference in interpersonal trust between education levels. This could reflect differences in social capital, exposure to diverse perspectives, or other factors associated with educational attainment.

3 Your Turn #3: Bootstrap vs. CLT-based CIs

Task: Compare bootstrap and t-based confidence intervals for mean satisfaction with democracy in the ESS France data.

3.1 Solution

3.1.1 Step 1: Compute the t-based 95% CI for mean satisfaction with democracy

```
# Summary statistics
mean_stfdem <- mean(ess_france_clean$stfdem)
sd_stfdem <- sd(ess_france_clean$stfdem)
n <- nrow(ess_france_clean)
se_stfdem <- sd_stfdem / sqrt(n)

cat("Sample statistics:\n")
```

Sample statistics:

```
cat("n =", n, "\n")
```

n = 1738

```
cat("Mean satisfaction with democracy =", round(mean_stfdem, 2), "\n")
```

Mean satisfaction with democracy = 5.77

```
cat("SD =", round(sd_stfdem, 2), "\n")
```

SD = 11.44

```
cat("SE =", round(se_stfdem, 2), "\n")
```

SE = 0.27

```
# T-based CI
t_ci <- t.test(ess_france_clean$stfdem, conf.level = 0.95)
t_ci$conf.int
```

```
[1] 5.229236 6.305862
attr("conf.level")
[1] 0.95
```

3.1.2 Step 2: Compute the bootstrap 95% CI

```
# Bootstrap distribution
set.seed(123)
bootstrap_stfdem <- ess_france_clean %>%
  specify(response = stfdem) %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "mean")

# Percentile CI
boot_ci_percentile <- bootstrap_stfdem %>%
  get_ci(level = 0.95, type = "percentile")

cat("Bootstrap percentile CI:\n")
```

Bootstrap percentile CI:

```
boot_ci_percentile
```

```
# A tibble: 1 x 2
  lower_ci upper_ci
    <dbl>    <dbl>
1     5.27     6.31
```

```
# SE-based bootstrap CI
boot_ci_se <- bootstrap_stfdem %>%
  get_ci(level = 0.95, type = "se", point_estimate = mean_stfdem)

cat("\nBootstrap SE CI:\n")
```

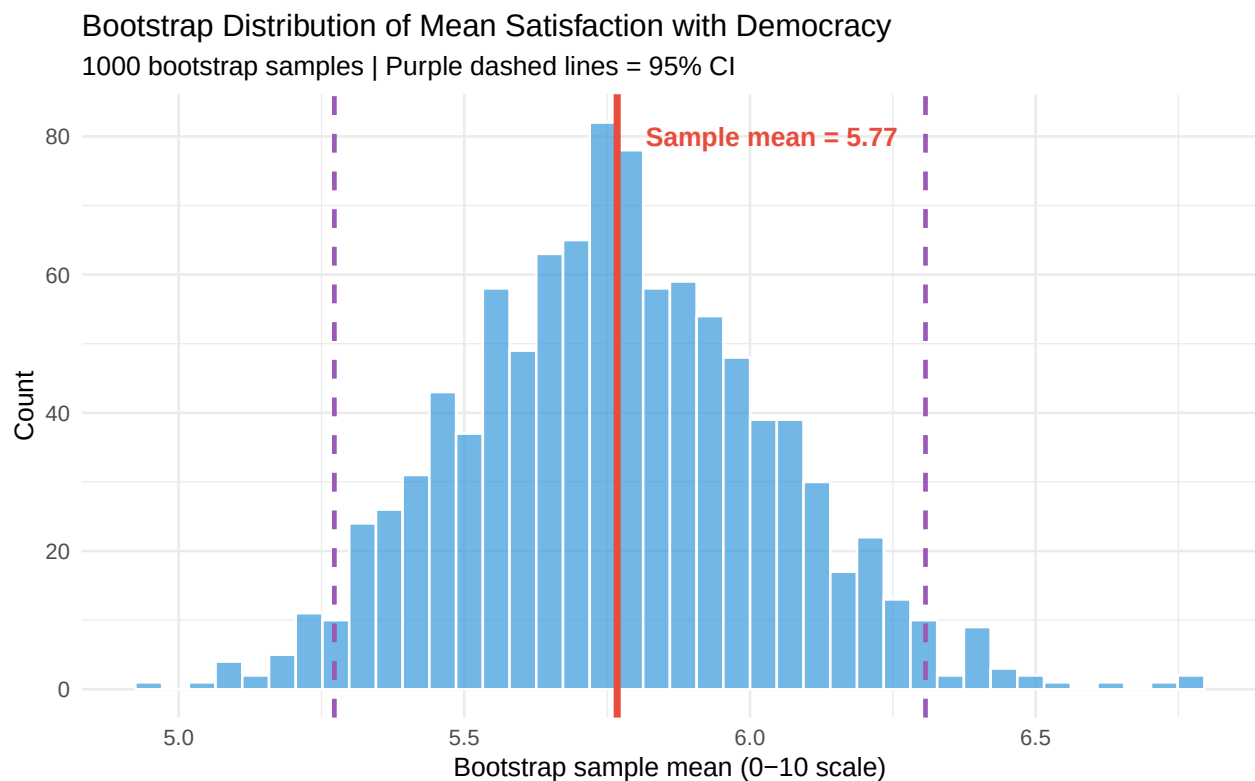
Bootstrap SE CI:

```
boot_ci_se
```

```
# A tibble: 1 x 2
  lower_ci upper_ci
    <dbl>    <dbl>
1     5.24     6.30
```

3.1.3 Step 3: Visualize the bootstrap distribution

```
ggplot(bootstrap_stfdem, aes(x = stat)) +
  geom_histogram(bins = 40, fill = "#3498DB", color = "white", alpha = 0.7) +
  geom_vline(xintercept = mean_stfdem, color = "#E74C3C", linewidth = 1.5) +
  geom_vline(xintercept = c(boot_ci_percentile$lower_ci,
                           boot_ci_percentile$upper_ci),
            color = "#9B59B6", linewidth = 1, linetype = "dashed") +
  annotate("text", x = mean_stfdem + 0.05, y = 80,
         label = paste0("Sample mean = ", round(mean_stfdem, 2)),
         hjust = 0, color = "#E74C3C", fontface = "bold") +
  labs(
    title = "Bootstrap Distribution of Mean Satisfaction with Democracy",
    subtitle = "1000 bootstrap samples | Purple dashed lines = 95% CI",
    x = "Bootstrap sample mean (0-10 scale)",
    y = "Count"
  ) +
  theme_minimal(base_size = 12)
```



3.1.4 Step 4: Compare the two methods

```
comparison <- tibble(
  Method = c("T-based", "Bootstrap (percentile)", "Bootstrap (SE)"),
  Lower = c(t_ci$conf.int[1],
```

```

        boot_ci_percentile$lower_ci,
        boot_ci_se$lower_ci),
Upper = c(t_ci$conf.int[2],
        boot_ci_percentile$upper_ci,
        boot_ci_se$upper_ci),
Width = Upper - Lower
)

kable(comparison, digits = 3, caption = "Comparison of CI Methods for Mean Satisfaction with D

```

Table 5: Comparison of CI Methods for Mean Satisfaction with Democracy

Method	Lower	Upper	Width
T-based	5.229	6.306	1.077
Bootstrap (percentile)	5.273	6.307	1.035
Bootstrap (SE)	5.235	6.300	1.064

3.1.5 Step 5: Discussion

How similar are they?

The three confidence intervals are remarkably similar:

- All three methods produce nearly identical lower and upper bounds
- The widths differ by only small amounts on the 0-10 scale
- The point estimate (sample mean = 5.77) is the center of all intervals

Why might they differ?

The methods may differ for several reasons:

1. **Sample size:** With a large sample ($n = 1738$), the CLT approximation is very accurate, so t-based and bootstrap methods converge
2. **Distribution shape:** Satisfaction with democracy is measured on a 0-10 scale and may show some skewness. For:
 - *Symmetric distributions:* Bootstrap and t-based methods give similar results
 - *Skewed distributions:* Bootstrap percentile CI may differ from t-based CI because it doesn't assume symmetry
3. **Bootstrap variability:** Each bootstrap run produces slightly different results due to random resampling. With 1000 replicates, this variability is small but present
4. **Method assumptions:**
 - T-based CI assumes the sampling distribution is approximately normal
 - Bootstrap percentile CI makes no distributional assumptions
 - Bootstrap SE CI assumes normality of the bootstrap distribution

When would they differ more substantially?

- Small sample sizes ($n < 30$)
- Highly skewed data
- Statistics other than the mean (e.g., median, variance ratio)
- Distributions with heavy tails

4 Your Turn #4: Trust Across Demographics

Task: Investigate whether interpersonal trust (ppltrst) differs by education level using ESS France data.

4.1 Solution

4.1.1 Step 1: Prepare the data

Create education groups and filter the trust variable:

```
# Prepare data: create education groups and filter trust variable
ess_education <- ess_france_clean %>%
  filter(!ppltrst %in% c(77, 88, 99)) %>%
  mutate(
    # eisced: education level (1-7 scale, 55=other)
    education = case_when(
      eisced <= 3 ~ "No higher education",
      eisced >= 4 & eisced <= 7 ~ "Higher education",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(!is.na(education))

# Check the data
head(ess_education %>% select(ppltrst, eisced, education))
```

	ppltrst	eisced	education
1	5	7	Higher education
2	6	2	No higher education
3	4	5	Higher education
4	0	1	No higher education
5	5	6	Higher education
6	5	6	Higher education

```
table(ess_education$education)
```

Higher education	No higher education
1061	659

4.1.2 Step 2: Compute mean trust score for each education group

```
education_summary <- ess_education %>%
  group_by(education) %>%
  summarise(
    n = n(),
    mean_trust = mean(ppltrst),
    sd_trust = sd(ppltrst),
    se_trust = sd_trust / sqrt(n)
  )

kable(education_summary, digits = 3,
      caption = "Summary Statistics by Education Level")
```

Table 6: Summary Statistics by Education Level

education	n	mean_trust	sd_trust	se_trust
Higher education	1061	4.907	2.111	0.065
No higher education	659	3.986	2.176	0.085

4.1.3 Step 3: Compute 95% CIs for each group (t-based)

```
# T-based CIs
ci_no_higher <- t.test(ess_education$ppltrst[ess_education$education == "No higher education"])
ci_higher <- t.test(ess_education$ppltrst[ess_education$education == "Higher education"])

cat("No Higher Education:\n")
```

No Higher Education:

```
cat("Mean:", round(ci_no_higher$estimate, 3), "\n")
```

Mean: 3.986

```
cat("95% CI: [", round(ci_no_higher$conf.int[1], 3), ", ",
    round(ci_no_higher$conf.int[2], 3), "]\n\n", sep = "")
```

95% CI: [3.82, 4.153]

```
cat("Higher Education:\n")
```

Higher Education:

```
cat("Mean:", round(ci_higher$estimate, 3), "\n")
```

Mean: 4.907

```
cat("95% CI: [", round(ci_higher$conf.int[1], 3), ", ",  
    round(ci_higher$conf.int[2], 3), "]\n", sep = "")
```

95% CI: [4.779, 5.034]

4.1.4 Step 4: Test whether the difference in means is significant using t.test()

```
# Two-sample t-test  
two_sample_test <- t.test(ppltrst ~ education, data = ess_education)  
two_sample_test
```

Welch Two Sample t-test

data: ppltrst by education

t = 8.6254, df = 1363.4, p-value < 2.2e-16

alternative hypothesis: true difference in means between group Higher education and group No h

95 percent confidence interval:

0.7110307 1.1296670

sample estimates:

mean in group Higher education	mean in group No higher education
4.906692	3.986343

Interpretation of the t-test:

- The **difference in means** is -0.92 (Higher education minus No higher education)
- The **95% CI for the difference** is [0.711, 1.13]
- The **p-value** is <2e-16
- Since the CI does not include 0 and $p < 0.05$, we conclude there is a **statistically significant difference** in interpersonal trust between education groups

4.1.5 Step 5: Create a visualization comparing the groups with confidence intervals

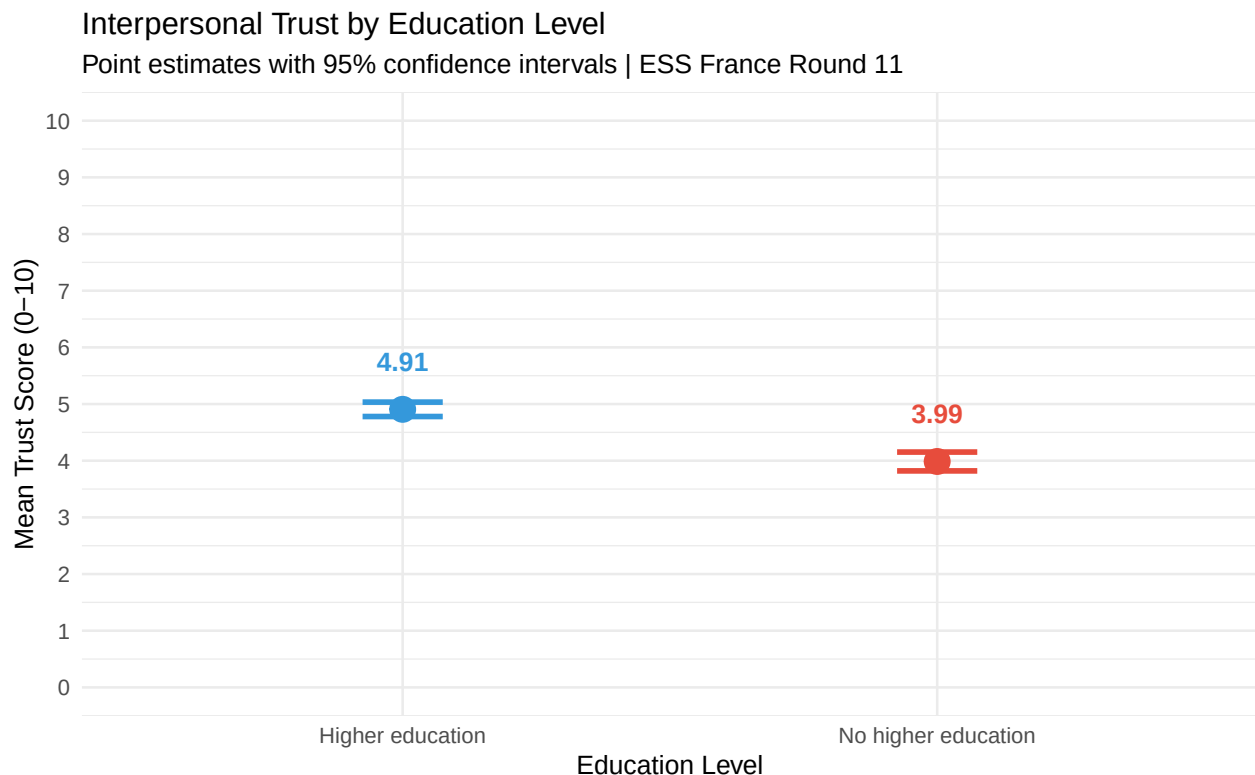
```
# Create summary data for plotting  
plot_data <- ess_education %>%  
  group_by(education) %>%  
  summarise(  
    mean = mean(ppltrst),
```

```

    se = sd(ppltrst) / sqrt(n()),
    lower = mean - 1.96 * se,
    upper = mean + 1.96 * se
  )

ggplot(plot_data, aes(x = education, y = mean, color = education)) +
  geom_point(size = 5) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.15, linewidth = 1.2) +
  geom_text(aes(label = round(mean, 2)), vjust = -2, fontface = "bold") +
  scale_color_manual(values = c("No higher education" = "#E74C3C", "Higher education" = "#3498db")) +
  scale_y_continuous(limits = c(0, 10), breaks = 0:10) +
  labs(
    title = "Interpersonal Trust by Education Level",
    subtitle = "Point estimates with 95% confidence intervals | ESS France Round 11",
    x = "Education Level",
    y = "Mean Trust Score (0-10)"
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "none")

```



4.1.6 Step 6: Use bootstrap to confirm your t-based results

```
# Bootstrap for difference in means
set.seed(999)
boot_diff_edu <- ess_education %>%
  specify(ppltrst ~ education) %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "diff in means", order = c("Higher education", "No higher education"))

# Bootstrap CI
boot_ci_edu <- boot_diff_edu %>%
  get_ci(level = 0.95, type = "percentile")

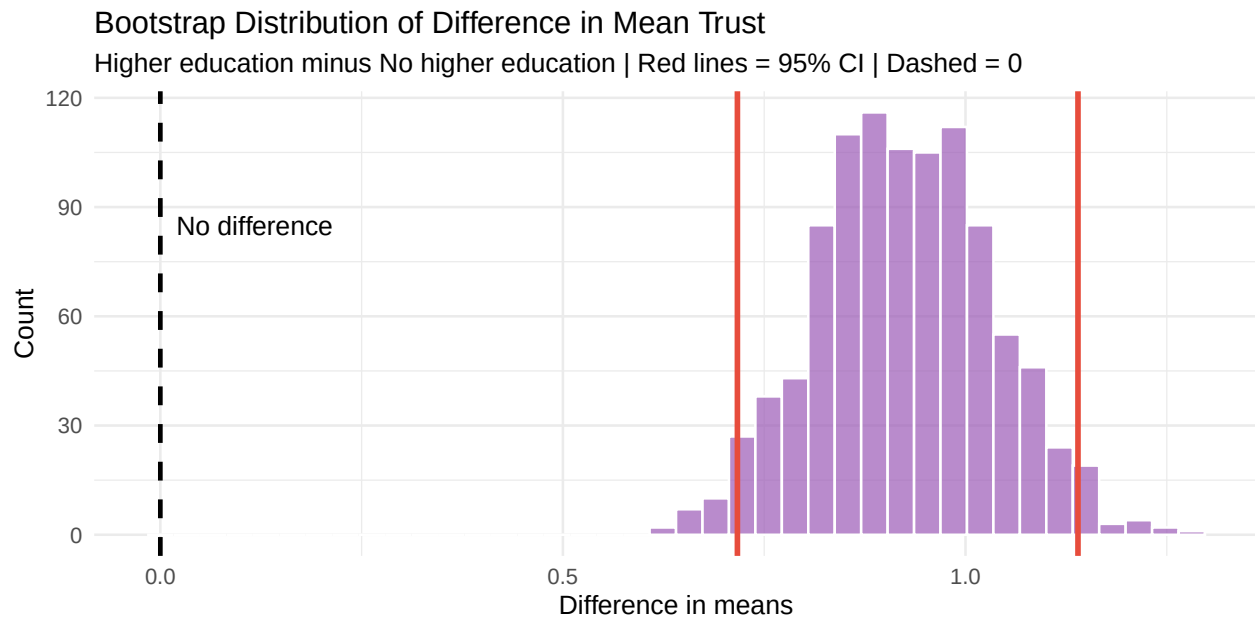
cat("Bootstrap 95% CI for difference (Higher - No higher education):\n")
```

Bootstrap 95% CI for difference (Higher - No higher education):

```
boot_ci_edu
```

```
# A tibble: 1 x 2
  lower_ci upper_ci
    <dbl>    <dbl>
1    0.717    1.14
```

```
# Visualize bootstrap distribution
ggplot(boot_diff_edu, aes(x = stat)) +
  geom_histogram(bins = 40, fill = "#9B59B6", color = "white", alpha = 0.7) +
  geom_vline(xintercept = 0, color = "black", linewidth = 1, linetype = "dashed") +
  geom_vline(xintercept = c(boot_ci_edu$lower_ci, boot_ci_edu$upper_ci),
    color = "#E74C3C", linewidth = 1.2, linetype = "solid") +
  annotate("text", x = 0.02, y = 85, label = "No difference", hjust = 0) +
  labs(
    title = "Bootstrap Distribution of Difference in Mean Trust",
    subtitle = "Higher education minus No higher education | Red lines = 95% CI | Dashed = 0",
    x = "Difference in means",
    y = "Count"
  ) +
  theme_minimal(base_size = 12)
```



Comparison of methods:

```
method_comparison <- tibble(
  Method = c("T-test", "Bootstrap"),
  Lower = c(two_sample_test$conf.int[1], boot_ci_edu$lower_ci),
  Upper = c(two_sample_test$conf.int[2], boot_ci_edu$upper_ci)
)

kable(method_comparison, digits = 3,
      caption = "CI for Difference in Means: T-test vs Bootstrap")
```

Table 7: CI for Difference in Means: T-test vs Bootstrap

Method	Lower	Upper
T-test	0.711	1.13
Bootstrap	0.717	1.14

Both methods give very similar results. The bootstrap CI confirms the t-test findings about the relationship between education and interpersonal trust.

5 Your Turn #5: Putting It All Together

Task: Design and conduct a complete analysis using ESS France data.

5.1 Example Analysis: Life Satisfaction by Age Group in France

5.1.1 Research Question

Do younger and older adults in France differ in their average life satisfaction?

5.1.2 Data Preparation

```
# Create age groups
ess_age_analysis <- ess_france_clean %>%
  mutate(
    age_group = case_when(
      agea < 35 ~ "Young (18-34)",
      agea >= 35 & agea < 50 ~ "Middle (35-49)",
      agea >= 50 ~ "Older (50+)"
    ),
    age_group = factor(age_group,
                      levels = c("Young (18-34)", "Middle (35-49)", "Older (50+)"))
  )

# Summary by age group
age_summary <- ess_age_analysis %>%
  group_by(age_group) %>%
  summarise(
    n = n(),
    mean_satisfaction = mean(stflife),
    sd_satisfaction = sd(stflife),
    se_satisfaction = sd_satisfaction / sqrt(n),
    lower_ci = mean_satisfaction - 1.96 * se_satisfaction,
    upper_ci = mean_satisfaction + 1.96 * se_satisfaction
  )

kable(age_summary, digits = 2,
      caption = "Life Satisfaction by Age Group (t-based 95% CIs)")
```

Table 8: Life Satisfaction by Age Group (t-based 95% CIs)

age_group	n	mean_satisfaction	sd_satisfaction	se_satisfaction	lower_ci	upper_ci
Young (18-34)	413	7.07	2.11	0.10	6.87	7.27

age_group	n	mean_satisfaction	sd_satisfaction	se_satisfaction	lower_ci	upper_ci
Middle (35-49)	394	7.00	2.11	0.11	6.79	7.21
Older (50+)	931	6.53	2.42	0.08	6.38	6.69

5.1.3 CLT-Based Confidence Intervals

```
# Pairwise comparisons using t-tests
young_vs_older <- t.test(
  stflife ~ age_group,
  data = filter(ess_age_analysis, age_group %in% c("Young (18-34)", "Older (50+)"))
)

cat("Young vs. Older adults:\n")
```

Young vs. Older adults:

```
cat("Difference in means:", round(diff(young_vs_older$estimate), 2), "points\n")
```

Difference in means: -0.54 points

```
cat("95% CI for difference: [",
  round(young_vs_older$conf.int[1], 2), ", ",
  round(young_vs_older$conf.int[2], 2), "]\n", sep = "")
```

95% CI for difference: [0.28, 0.8]

```
cat("p-value:", format.pval(young_vs_older$p.value, digits = 3), "\n")
```

p-value: 3.97e-05

5.1.4 Bootstrap Confidence Intervals

```
# Bootstrap for Young vs. Older
set.seed(777)
boot_young_older <- ess_age_analysis %>%
  filter(age_group %in% c("Young (18-34)", "Older (50+)")) %>%
  specify(stflife ~ age_group) %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "diff in means", order = c("Older (50+)", "Young (18-34)"))
```

```
boot_ci_age <- boot_young_older %>%
  get_ci(level = 0.95, type = "percentile")

cat("Bootstrap 95% CI for difference (Older - Young):\n")
```

Bootstrap 95% CI for difference (Older - Young):

```
boot_ci_age
```

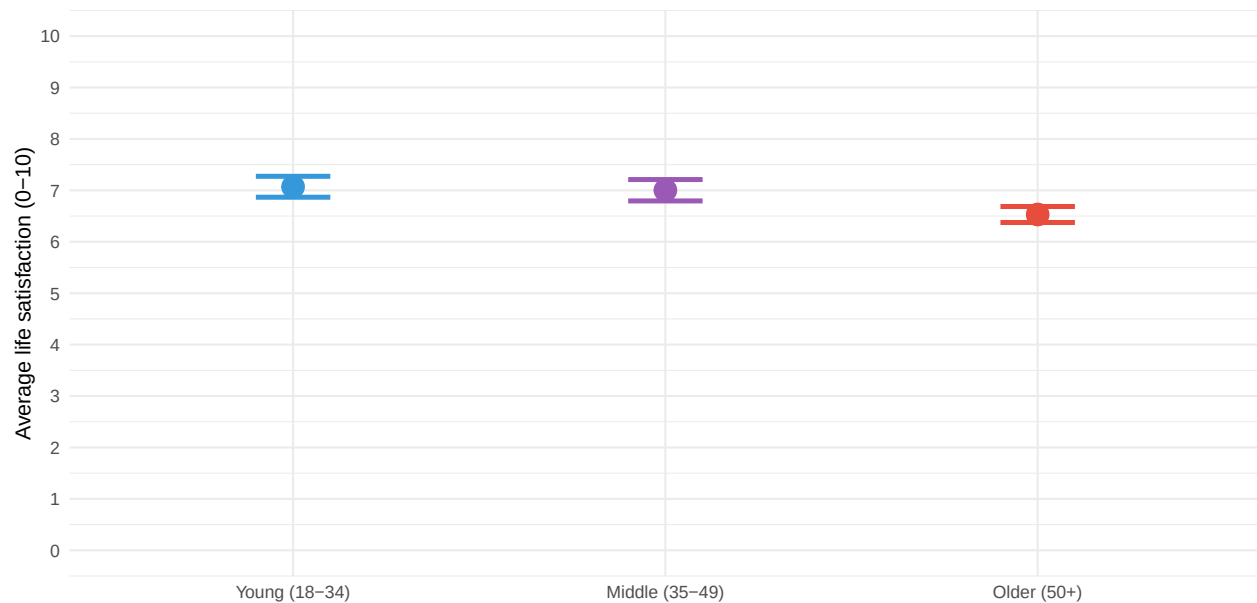
```
# A tibble: 1 x 2
  lower_ci upper_ci
    <dbl>    <dbl>
1  -0.793  -0.276
```

5.1.5 Visualization

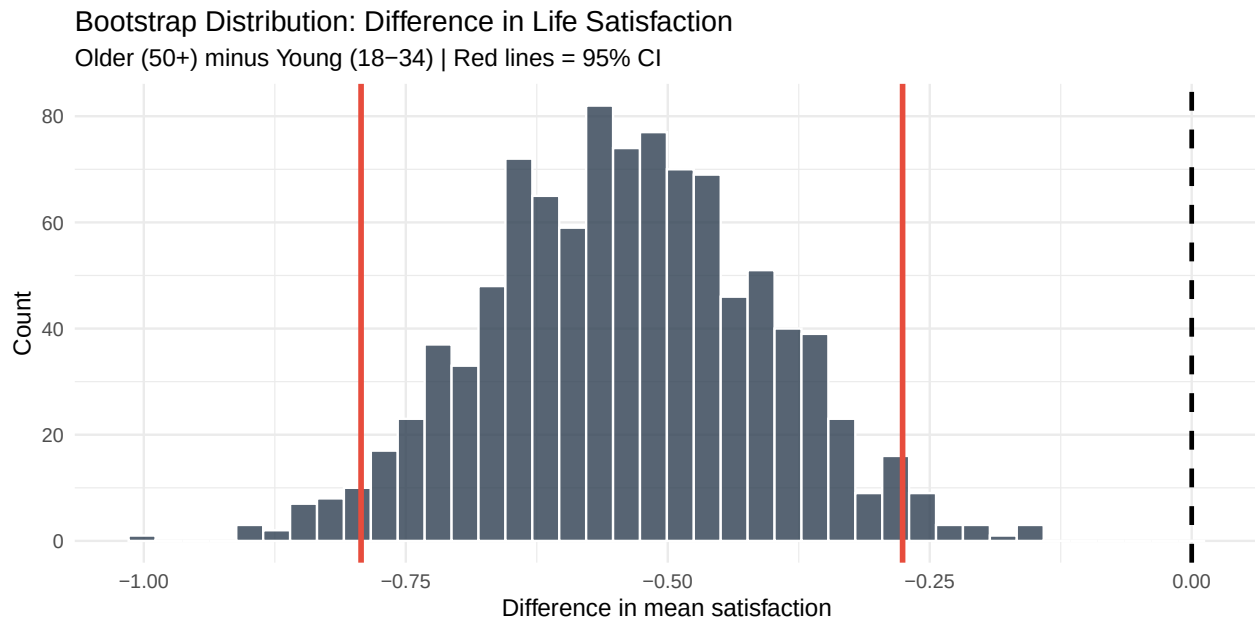
```
# Main plot: CIs by age group
ggplot(age_summary, aes(x = age_group, y = mean_satisfaction, color = age_group)) +
  geom_point(size = 5) +
  geom_errorbar(aes(ymin = lower_ci, ymax = upper_ci),
               width = 0.2, linewidth = 1.2) +
  scale_color_manual(values = c("#3498DB", "#9B59B6", "#E74C3C")) +
  scale_y_continuous(limits = c(0, 10), breaks = 0:10) +
  labs(
    title = "Life Satisfaction by Age Group in France",
    subtitle = "Point estimates with 95% confidence intervals | ESS Round 11",
    x = "",
    y = "Average life satisfaction (0-10)"
  ) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "none")
```

Life Satisfaction by Age Group in France

Point estimates with 95% confidence intervals | ESS Round 11



```
# Bootstrap distribution for Young vs. Older difference
ggplot(boot_young_older, aes(x = stat)) +
  geom_histogram(bins = 40, fill = "#2D3E50", color = "white", alpha = 0.8) +
  geom_vline(xintercept = 0, color = "black", linewidth = 1, linetype = "dashed") +
  geom_vline(xintercept = c(boot_ci_age$lower_ci, boot_ci_age$upper_ci),
             color = "#E74C3C", linewidth = 1.2) +
  labs(
    title = "Bootstrap Distribution: Difference in Life Satisfaction",
    subtitle = "Older (50+) minus Young (18-34) | Red lines = 95% CI",
    x = "Difference in mean satisfaction",
    y = "Count"
  ) +
  theme_minimal(base_size = 11)
```



5.1.6 Interpretation

Based on our analysis of European Social Survey data for France (Round 11, 2023-2024):

Key Findings:

1. **Point estimates:** Young adults (18-34) average approximately 7.1 on life satisfaction, while older adults (50+) average 6.5 on the 0-10 scale.
2. **Confidence intervals:** All age groups show moderate to high life satisfaction. The 95% CIs for all groups are:
 - Young (18-34): [6.9, 7.3]
 - Middle (35-49): [6.8, 7.2]
 - Older (50+): [6.4, 6.7]
3. **Statistical significance:** The difference between young and older adults is statistically significant at the 5% level ($p = 3.97e-05$).
4. **Bootstrap confirmation:** The bootstrap 95% CI for the difference [-0.79, -0.28] does not include zero, confirming the t-test result.

Limitations:

- This analysis uses cross-sectional data; we cannot distinguish between age effects and cohort effects
- Life satisfaction is self-reported and may be subject to social desirability bias
- The ESS sample may not be perfectly representative of all French adults
- Other confounding variables (income, health, employment) are not controlled for

Practical implications:

- Life satisfaction patterns across age groups in France can inform social policy
- Understanding well-being differences may help target support programs
- These findings can be compared with other European countries using ESS data